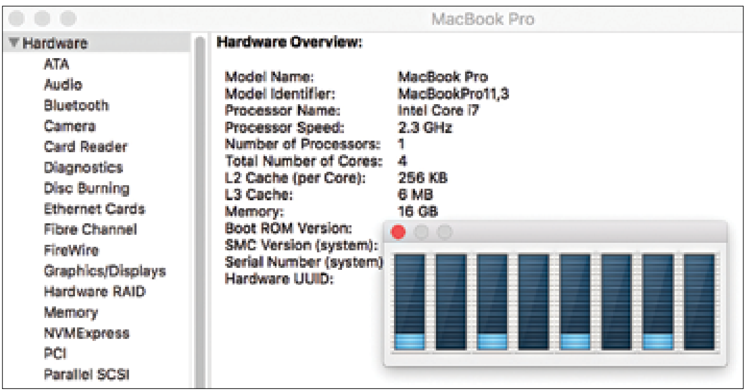
in the picture). This is because there are only 4 actual compute cores available but to make sure that cache misses have as less an impact as possible, modern processors use SMT (Simultaneous Multi-Threading, HyperThreading in Intel's terminology) and keep 2 states for every compute core. This way when the processor is waiting for some other resource (e.g. while a cache miss is being dealt with), the other state can be resumed on the core. This way, a Hyper-Threaded CPU mentioned as example will perform better in a multi-tasking environment but you cannot expect it to work just as fast as another CPU with 8 physical cores under most conditions. If, however, you put a more compute-intensive and less memory-intensive job, the performance will not be the same as you would expect from an octa-core.



Notice that the number of cores mentioned in the specification is 4 while activity monitor on the Mac shows 8 cores being used.

Having said all of that, you should be able to recognize the ones meant for you by their names. The older Core Duo/Core 2 Duo series from Intel were meant for the desktop and so are the new i3/i5/i7 series processors. AMD on the other hand, has the FX series, APUs and the new RYZEN 3/5/7 chips for desktops. At the same time, if the packaging reads Xeon or EPYC, then you should stay away from them unless you know what you are doing – they are meant for the servers and are both extremely powerful as well as pricey! They can have up to 32 cores which can run up to 64 threads. They also have a higher cache/core ratio than desktop lineups. However, if you want such a beast to live in your house though, who are we to complain?